SINGAPORE SCHOOL KINDERLAND

SECONDARY-1 (SEC 2) • COMPUTER SCIENCE / IT

FILE SIZE CALCULATIONS — MASTER FORMULA GUIDE & 20-QUESTION WORKSHEET

PART 1: COMPLETE FORMULAS, UNITS & SHORTCUT TRICKS

1. Storage Units: 4 bits = 1 Nibble | 8 bits = 1 Byte | 1,024 Bytes = 1 KiB | 1,000 Bytes = 1 KB | 1,024 KiB = 1 MiB | 1,000 KB = 1 MB.

2. Frequency & Time Units: 1 kHz = 1,000 Hz | 1 MHz = 1,000,000 Hz | 1 Minute = 60 Seconds | Mono = 1 Channel | Stereo = 2 Channels.

3. Text File Size Formula: Number of Characters × Bits per Character (8-bit ASCII = 1 Byte/char; 16-bit Unicode = 2 Bytes/char).

4. Bitmap Image Size Formula: Width (pixels) × Height (pixels) × Colour Depth (bits) ÷ 8 = Size in Bytes.

Colour Depth Trick: 2 colours = 1-bit | 4 colours = 2-bit | 16 colours = 4-bit | 256 colours = 8-bit (1B) | 65,536 colours = 16-bit (2B) | 16.7M colours = 24-bit (3B).

5. Digital Sound Size Formula: Sample Rate (Hz) × Sample Resolution (bits) × Time (seconds) × Channels ÷ 8 = Size in Bytes.

Section A: Unit Conversions (Hz, kHz, MHz, KB/KiB, MB/MiB)

Question 1 [3 Marks]: Convert the following frequency and sample rate values: (a) 8 kHz into Hz (b) 44.1 kHz into Hz (c) 2.5 MHz into kHz and Hz

WORKING & ANSWER BOX:

Final Answer: _____

Question 2 [3 Marks]: Convert the following binary data measurements: (a) 64 bits into Bytes (b) 400 Bytes into bits (c) 8,192 bits into Bytes, and then into KiB (Kibibytes)

WORKING & ANSWER BOX:

Final Answer: _____

Question 3 [3 Marks]: Convert the following storage capacity units: (a) 5 KiB into Bytes (using binary factor 1,024) (b) 3 MB into Bytes (using decimal factor 1,000,000) (c) 2,048 KiB into MiB

WORKING & ANSWER BOX:

Final Answer: _____

Question 4 [2 Marks]: Convert the following time durations and audio channel settings for sound calculations: (a) 2.5 minutes into seconds (b) State the number of channels for Mono audio and Stereo audio.

WORKING & ANSWER BOX:

Final Answer: _____

Section B: Text File Size Calculations (ASCII & Unicode)

Question 5 [3 Marks]: A student types the following phrase in a plain text editor using 8-bit ASCII encoding: "HELLO SECONDARY 1" Count the total number of characters (including spaces) and calculate the file size in bits and Bytes.

WORKING & ANSWER BOX:

Final Answer: _____

Question 6 [3 Marks]: A plain text essay contains 4,096 characters (including spaces and punctuation). It is saved using standard 8-bit ASCII encoding. Calculate the size of the text file in: (a) Bytes (b) KiB (Kibibytes)

WORKING & ANSWER BOX:

Final Answer: _____

Question 7 [3 Marks]: A multi-language webpage contains 3,500 characters and is saved using 16-bit Unicode encoding. Calculate the file size in: (a) bits (b) Bytes (c) KB (Kilobytes, dividing by 1,000)

WORKING & ANSWER BOX:

Final Answer: _____

Question 8 [4 Marks]: A digital e-book has 10 pages. Each page has 32 lines, and each line has 64 characters (including spaces). The e-book uses 8-bit ASCII. Calculate the total file size of the 10-page e-book in Bytes and in KiB.

WORKING & ANSWER BOX:

Final Answer: _____

Section C: Bitmap Image File Size Calculations

Question 9 [3 Marks]: A small game icon has a resolution of 100 × 80 pixels and uses a colour depth of 8 bits per pixel. Calculate the uncompressed image file size in: (a) bits (b) Bytes (c) KB (dividing by 1,000)

WORKING & ANSWER BOX:

Final Answer: _____

Question 10 [3 Marks]: A black-and-white barcode image (2 colours only) has dimensions of 400 pixels wide by 200 pixels high. (a) State the colour depth in bits for a 2-colour image. (b) Calculate the file size of the barcode image in Bytes and KB.

WORKING & ANSWER BOX:

Final Answer: _____

Question 11 [4 Marks]: A retro arcade graphic has a resolution of 512 × 256 pixels and displays a maximum of 16 colours. (a) How many bits per pixel (colour depth) are required for 16 colours? (b) Calculate the exact file size of the graphic in KiB (dividing by 1,024).

WORKING & ANSWER BOX:

Final Answer: _____

Question 12 [3 Marks]: A high-quality digital photograph has a resolution of 1,000 × 800 pixels and uses 24-bit True Colour. Calculate the file size of the photograph in: (a) Bytes (b) MB (Megabytes, dividing by 1,000,000)

WORKING & ANSWER BOX:

Final Answer: _____

Question 13 [3 Marks]: A web banner has dimensions of 1,024 × 512 pixels and uses a 256-colour palette. (a) Determine the colour depth in bits and Bytes for 256 colours. (b) Calculate the file size of the banner in KiB.

WORKING & ANSWER BOX:

Final Answer: _____

Question 14 [4 Marks]: A designer saves 5 uncompressed images on a flash drive. Each image has a resolution of 2,000 × 1,000 pixels and a 16-bit colour depth. Calculate the total storage space required for all 5 images in MB (dividing by 1,000,000).

WORKING & ANSWER BOX:

Final Answer: _____

Section D: Digital Sound File Size Calculations

Question 15 [3 Marks]: A 10-second mono voice note is recorded with a sample rate of 8,000 Hz and a sample resolution of 8 bits. Calculate the file size in: (a) bits (b) Bytes (c) KB (dividing by 1,000)

WORKING & ANSWER BOX:

Final Answer: _____

Question 16 [4 Marks]: A 30-second stereo sound clip is recorded at a sample rate of 20 kHz with a sample resolution of 16 bits. Calculate the file size of the audio clip in: (a) Bytes (b) MB (dividing by 1,000,000)

WORKING & ANSWER BOX:

Final Answer: _____

Question 17 [4 Marks]: A 1-minute mono podcast is recorded at a sample rate of 10 kHz and a sample resolution of 16 bits. Calculate the total file size in Bytes and in KB (dividing by 1,000).

WORKING & ANSWER BOX:

Final Answer: _____

Question 18 [4 Marks]: A 20-second stereo song is recorded at CD quality with a sample rate of 44.1 kHz and a sample resolution of 16 bits. Calculate the uncompressed audio file size in Bytes and MB (dividing by 1,000,000).

WORKING & ANSWER BOX:

Final Answer: _____

Question 19 [4 Marks]: A 2-minute stereo background music track is sampled at 16,000 Hz with an 8-bit resolution. Calculate the file size in Bytes and in MB (dividing by 1,000,000).

WORKING & ANSWER BOX:

Final Answer: _____

Section E: Master Multimedia Challenge

Question 20 [5 Marks]: A student creates a multimedia slide containing three elements: 1. Text: 2,000 characters encoded in 8-bit ASCII. 2. Image: One bitmap picture of 500 × 400 pixels with 8-bit colour depth. 3. Audio: One 5-second mono voice clip sampled at 10 kHz with 8-bit resolution. Calculate the combined total file size of all three elements in KB (dividing by 1,000).

WORKING & ANSWER BOX:

Final Answer: _____